IBM

Cloud Insights

NetApp
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IBM

IBM Cleversafe data collector

Cloud Insights uses this data collector to discover inventory and performance data for IBM Cleversafe storage systems.

Terminology

Cloud Insights acquires the following inventory information from the IBM Cleversafe data collector. For each asset type acquired by Cloud Insights, the most common terminology used for this asset is shown. When viewing or troubleshooting this data collector, keep the following terminology in mind:

Vendor/Model Term	Cloud Insights Term
Storage Pool	Storage Pool
Container	Internal Volume
Container	File Share
NFS Share	Share

Note: These are common terminology mappings only and might not represent every case for this data collector.

Requirements

- The external data services IP address for the cluster
- Administrator user name and password
- Port 9440

Configuration

Field	Description
Manager IP or host name	IP address or hostname of management node
User name	Username for the user account with super user or system administrator role
Password	Password for the user account with super user or system administrator role

Advanced configuration

Field	Description
Inventory poll interval (min)	Interval between inventory polls.
HTTP Connection Timeout (sec)	HTTP timeout in seconds.

Troubleshooting

Additional information on this Data Collector may be found from the Support page or in the Data Collector Support Matrix.

IBM CS data collector

Cloud Insights uses this data collector to discover inventory and performance data for IBM CS storage systems.

Terminology

Cloud Insights acquires the following inventory information from the IBM CS data collector. For each asset type acquired by Cloud Insights, the most common terminology used for this asset is shown. When viewing or troubleshooting this data collector, keep the following terminology in mind:

Vendor/Model Term	Cloud Insights Term
Storage Pool	Storage Pool
Container	Internal Volume
Container	File Share
NFS Share	Share

Note: These are common terminology mappings only and might not represent every case for this data collector.

Requirements

- The external data services IP address for the cluster
- Administrator user name and password
- Port 9440

Configuration

Field	Description
Prism External IP Address	The external data services IP address for the cluster
User name	User name for the Admin account
Password	Password for the Admin account

Advanced configuration

Field	Description
TCP port	TCP Port used to connect to the IBM CS array. The default is 9440.
Inventory poll interval (min)	Interval between inventory polls. The default is 60 minutes.

Field	Description
Performance poll interval(sec)	Interval between performance polls. The default is 300 seconds.

Troubleshooting

Additional information on this Data Collector may be found from the Support page or in the Data Collector Support Matrix.

IBM System Storage DS8000 Series data collector

The IBM DS (CLI) data collector supports inventory and performance data acquisition for DS6xxx and DS8xxx devices.

DS3xxx, DS4xxx, and DS5xxx devices are supported by the NetApp E-Series data collector. You should refer to the Cloud Insights support matrix for supported models and firmware versions.

Terminology

Cloud Insights acquires the following inventory information from the IBM DS data collector. For each asset type acquired by Cloud Insights, the most common terminology used for this asset is shown. When viewing or troubleshooting this data collector, keep the following terminology in mind:

Vendor/Model Term	Cloud Insights Term
Disk Drive Module	Disk
Storage Image	Storage
Extent Pool	Storage Node
Fixed Block Volume	Volume
Host FC Initiator (Mapped)	Volume Mask

Note: These are common terminology mappings only and might not represent every case for this data collecor.

Requirements

You need the following to configure this data collector:

- IP address of each DS array
- Read-only username and password on each DS array
- Third-party software installed on the Cloud Insights AU: IBM *dscli*
- Access validation: Run *dscli* commands using the username and password
- Port requirements: 80, 443, & 1750

Configuration

Field	Description
DS Storage	IP address or fully-qualified domain name of the DS device
User Name	User name for the DS CLI
Password	Password for the DS CLI
dscli executable path	Full path to the *dscli* executable

Advanced configuration

Field	Description
Inventory Poll Interval (min)	Interval between inventory polls (min). The default is 40.
Storage Display Name	Name of the IBM DS storage array
Inventory Exclude Devices	Comma-separated list of device serial numbers to exclude from inventory collection
Performance Poll Interval (sec)	The default is 300.
Performance Filter Type	Include: Data collected only from devices on list. Exclude: No data from these devices is collected
Performance Filter Device List	Comma-separated list of device IDs to include or exclude from performance collection

Troubleshooting

Some things to try if you encounter problems with this data collector:

Inventory

Problem:	Try this:
Error containing: CMUC00192E, CMUC00191E or CMUC00190E	* Verify credentials and IP address entered. * Try to communicate with the array through web management console https://${ip}:8452/DS8000/Console. Replace the ${ip} with data collector configured IP.
Error: * Cannot run program * Error executing command	* From Cloud Insights Acquisition Unit Open a CMD * Open CLI.CFG file in CLI's home dir/lib and check property JAVA_INSTALL, edit the value to match your environment * Display Java version installed on this machine, typing: "java -version" * Ping the IP address of the IBM Storage device specified in CLI command issued. * If all the above worked fine then manually run a CLI command

Additional information may be found from the Support page or in the Data Collector Support Matrix.

Configuring the IBM PowerVM data collector

The IBM PowerVM (SSH) data collector is used to collect information about virtual partitions running on IBM POWER hardware instances managed by a hardware management console (HMC).

Terminology

Cloud Insights acquires inventory information from the virtual partitions running on IBM POWER hardware instances. For each asset type acquired, the most common terminology used for the asset is shown. When viewing or troubleshooting this data collector, keep the following terminology in mind:

Vendor/Model Term	Cloud Insights Term
hdisk	Virtual Disk
Managed System	Host
LPAR, VIO Server	Virtual Machine
Volume Group	Data Store
Physical Volume	LUN

Note: These are common terminology mappings only and might not represent every case for this data collector.

Requirements

The following requirements must be met to configure and use this data collector:

- IP address of the Hardware Management Console (HMC)
- User name and password that provide access to Hardware Management Console (HMC) through SSH
- Port requirement SSH-22
- View permission on all management systems and logical partition security domains

  The user must also have View permission on HMC configurations and the ability to collect VPD information for the HMC console security grouping. The user must also be allowed Virtual IO Server Command access under the Logical Partition security grouping. It is a best practice to start from a role of an operator and then remove all roles. Read-only users on the HMC do not have privileges to run proxied commands on AIX hosts.

- IBM best practice is to have the devices monitored by two or more HMCs. Be aware that this may cause OnCommand Insight to report duplicated devices, therefore it is highly recommended to add redundant devices to the "Exclude Devices" list in the Advanced Configuration for this data collector.

Configuration

Field	Description
Hardware Management Console (HMC) IP Address	IP address or fully-qualified domain name of the PowerVM Hardware Management Console

Field	Description
HMC User	User name for the Hardware Management Console
Password	Password used for the Hardware Management Console

Advanced configuration

Field	Description
Inventory Poll Interval (min)	Interval between inventory polls. The default is 20 minutes.
SSH Port	Port used for SSH to the PowerVM
Password	Password used for the Hardware Management Console
Number of Retries	Number of inventory retry attempts
Exclude Devices	Comma-separated list of device IDs or display names to exclude

Troubleshooting

Additional information on this Data Collector may be found from the Support page or in the Data Collector Support Matrix.

Configuring the IBM SAN Volume Controller data collector

The IBM SAN Volume Controller (SVC) data collector collects inventory and performance data using SSH, supporting a variety of devices that run the SVC operating system.

The list of supported devices includes models such as the SVC, the v7000, the v5000, and the v3700. Refer to the Cloud Insights support matrix for supported models and firmware versions.

Terminology

Cloud Insights acquires the following inventory information from the IBM SVC data collector. For each asset type acquired by Cloud Insights, the most common terminology used for this asset is shown. When viewing or troubleshooting this data collector, keep the following terminology in mind:

Vendor/Model Term	Cloud Insights Term
Drive	Disk
Cluster	Storage
Node	Storage Node
Mdisk Group	Storage Pool
Vdisk	Volume
Mdisk	Backend LUNs and paths

Note: These are common terminology mappings only and might not represent every case for this data collector.

Inventory Requirements

- IP address of each SVC cluster
- Port 22 available
- Read-only user name and password

Performance Requirements

- SVC Console, which is mandatory for any SVC cluster and required for the SVC discovery foundation package.
- Credentials will require administrative access level only for copying performance files from cluster nodes to the config node.
- Enable data collection by connecting to the SVC cluster by SSH and running: *svctask startstats -interval 1*

  Note: Alternatively, enable data collection using the SVC management user interface.

Configuration

Field	Description
Cluster IP Addresses	IP addresses or fully-qualified domain names of the SVC storage
Inventory User Name	User name for the SVC CLI
Inventory Password	Password for the SVC CLI

Advanced configuration

Field	Description
Inventory Poll Interval (min)	Interval between inventory polls. The default is 40 minutes.
Performance Poll Interval (sec)	Interval between performance polls. The default is 300 seconds.
To clean up dumped stats files	Select this checkbox to clean up dumped stats files

Troubleshooting

Some things to try if you encounter problems with this data collector:

Problem:	Try this:
Error: "The command cannot be initiated because it was not run on the configuration node."	The command must be executed on the configuration node.

Some things to try if you encounter problems with this data collector:

Problem:	Try this:
Error: "The command cannot be initiated because it was not run on the configuration node."	The command must be executed on the configuration node.

Additional information on this Data Collector may be found from the Support page or in the Data Collector Support Matrix.

Configuring the IBM XIV/A9000 data collector

IBM XIV and A9000 (CLI) data collector uses the XIV command-line interface to collect inventory data while performance collection is accomplished by making SMI-S calls to the XIV/A9000 array, which runs a SMI-S provider on port 7778.

Terminology

Vendor/Model Term	Cloud Insights Term
Disk	Disk
Storage System	Storage
Storage Pool	Storage Pool
Volume	Volume

Requirements

The following requirements must be met to configure and use this data collector:

- Port requirement: TCP port 7778
- Read-only user name and password
- The XIV CLI must be installed on the AU

Performance requirements

The following are requirements for performance collection:

- SMI-S Agent 1.4 or higher
- SMI-S compatible CIMService running on array. Most XIV arrays have a CIMServer installed by default.
- User login must be provided for the CIMServer. The login must have full read access to the array configuration and properties.
- SMI-S namespace. Default is root/ibm. This is configurable in the CIMServer.
- Port Requirements: 5988 for HTTP, 5989 for HTTPS.
- Refer to the following link on how to create an account for SMI-S performance collection: http://publib.boulder.ibm.com/infocenter/tivihelp/v4r1/index.jsp?topic=%2Fcom.ibm.tpc_V41.doc%2Ffqz0_t_adding_cim_agent.html

Configuration

Field	Description
XIV IP address	IP address or fully-qualified domain name of the XIV storage
User Name	User name for the XIV storage
Password	Password for the XIV storage
Full Path to XIV CLI Directory	Full path to the folder containing the XIV CLI
SMI-S Host IP Address	IP address of the SMI-S host

Advanced configuration

Field	Description
Inventory Poll Interval (min)	Interval between inventory polls. The default is 40 minutes.
SMI-S Protocol	Protocol used to connect to the SMI-S provider. Also displays the default port.
Override SMI-S Port	If blank, use the default port in the Connection Type field, otherwise enter the connection port to use
Username	User name for the SMI-S Provider Host
Password	Password for the SMI-S Provider Host
Performance Poll Interval (sec)	Interval between performance polls. The default is 300 seconds.

Troubleshooting

Additional information on this Data Collector may be found from the Support page or in the Data Collector Support Matrix.

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